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CREATE DATABASE Projects;

USE Projects;

```
CREATE TABLE Employee (  
    empNo VARCHAR(5) PRIMARY KEY,  
    fName VARCHAR(20),  
    lName VARCHAR(20),  
    address VARCHAR(30),  
    DOB DATE,  
    sex VARCHAR(10),  
    position VARCHAR(15),  
    deptNo VARCHAR(5),  
  
    CONSTRAINT chk_sex CHECK(sex IN ("M", "F")),  
    CONSTRAINT chk_position CHECK(position IN ("Manager", "Team  
Leader", "Analyst", "Software Developer"))  
);
```

```
CREATE TABLE Department (  
    deptNo VARCHAR(5) PRIMARY KEY,  
    deptName VARCHAR(20),  
    mgrEmpNo VARCHAR (5)  
);
```

```
CREATE TABLE Project (  
    projNo VARCHAR(5) PRIMARY KEY,  
    projName VARCHAR(20),  
    deptNo VARCHAR(5),  
  
    CONSTRAINT fk_deptNo  
    FOREIGN KEY (deptNo)  
    REFERENCES Department(deptNo)  
);
```

```
CREATE TABLE WorksOn (  
    empNo VARCHAR(5),  
    projNo VARCHAR(5),  
    dateWorked DATE,  
    hoursWorked INT(3),  
  
    CONSTRAINT pkWorksOn  
    PRIMARY KEY(empNo, projNo, dateWorked),
```

```

CONSTRAINT fk_empNo
FOREIGN KEY (empNo)
REFERENCES Employee(empNo),

CONSTRAINT fk_projNo
FOREIGN KEY (projNo)
REFERENCES Project(projNo)
);

ALTER TABLE Employee
ADD CONSTRAINT fk_EmpDept
FOREIGN KEY (deptNo)
REFERENCES Department(deptNo);

ALTER TABLE WorksOn
ADD CONSTRAINT chk_hoursW CHECK(hoursWorked >0 AND hoursWorked <=
40);

ALTER TABLE Department
ADD CONSTRAINT fk_mgEmpNo
FOREIGN KEY (mgrEmpNo)
REFERENCES Employee(empNo);

SET FOREIGN_KEY_CHECKS = 0;

INSERT INTO Department (deptNo, deptName, mgrEmpNo) VALUES
('D0001', 'Frontend Dev', 'E0001'),
('D0002', 'Backend Dev', 'E0002'),
('D0003', 'Database Admin', 'E0003'),
('D0004', 'QA Testing', 'E0001'),
('D0005', 'UI/UX Design', 'E0002'),
('D0006', 'Cloud Ops', 'E0003'),
('D0007', 'Data Science', 'E0001'),
('D0008', 'Cybersecurity', 'E0002'),
('D0009', 'Tech Support', 'E0003'),
('D0010', 'R&D', 'E0001');

SELECT * FROM Department;

+-----+-----+-----+
| deptNo | deptName   | mgrEmpNo |
+-----+-----+-----+
| D0001 | Frontend Dev | E0001   |
| D0002 | Backend Dev  | E0002   |
| D0003 | Database Admin | E0003   |
| D0004 | QA Testing   | E0001   |
| D0005 | UI/UX Design | E0002   |

```

```

| D0006 | Cloud Ops    | E0003 |
| D0007 | Data Science | E0001 |
| D0008 | Cybersecurity | E0002 |
| D0009 | Tech Support | E0003 |
| D0010 | R&D          | E0001 |
+-----+-----+-----+

```

```

INSERT INTO Employee (empNo, fName, lName, address, DOB, sex, position,
deptNo) VALUES
('E0001', 'Alice', 'Smith', '123 Maple St', '1980-01-10', 'F', 'Manager', 'D0001'),
('E0002', 'Bob', 'Jones', '456 Oak St', '1982-02-20', 'M', 'Manager', 'D0002'),
('E0003', 'Charlie', 'Brown', '789 Pine St', '1985-03-30', 'M', 'Manager',
'D0003'),
('E0004', 'Diana', 'Prince', '111 Elm St', '1990-04-15', 'F', 'Leader', 'D0001'),
('E0005', 'Evan', 'Wright', '222 Cedar St', '1992-05-25', 'M', 'Leader', 'D0002'),
('E0006', 'Fiona', 'Gallagher', '333 Birch St', '1994-06-05', 'F', 'Analyst',
'D0003'),
('E0007', 'George', 'Clark', '444 Ash St', '1995-07-12', 'M', 'Analyst', 'D0004'),
('E0008', 'Hannah', 'Abbott', '555 Walnut St', '1996-08-22', 'F', 'Developer',
'D0001'),
('E0009', 'Ian', 'Malcolm', '666 Cherry St', '1997-09-09', 'M', 'Developer',
'D0002'),
('E0010', 'Julia', 'Stiles', '777 Spruce St', '1998-10-31', 'F', 'Developer', 'D0003');

SELECT * FROM Employee;

```

```

+-----+-----+-----+-----+-----+-----+-----+
+-----+
| empNo | fName | lName | address | DOB | sex | position | deptNo |
+-----+-----+-----+-----+-----+-----+-----+
+-----+
| E0001 | Alice | Smith | 123 Maple St | 1980-01-10 | F | Manager | D0001 |
| E0002 | Bob | Jones | 456 Oak St | 1982-02-20 | M | Manager | D0002 |
|
| E0003 | Charlie | Brown | 789 Pine St | 1985-03-30 | M | Manager |
D0003 |
| E0004 | Diana | Prince | 111 Elm St | 1990-04-15 | F | Leader | D0001 |
| E0005 | Evan | Wright | 222 Cedar St | 1992-05-25 | M | Leader | D0002 |
|
| E0006 | Fiona | Gallagher | 333 Birch St | 1994-06-05 | F | Analyst | D0003 |
|
| E0007 | George | Clark | 444 Ash St | 1995-07-12 | M | Analyst | D0004 |
|
| E0008 | Hannah | Abbott | 555 Walnut St | 1996-08-22 | F | Developer |
D0001 |
| E0009 | Ian | Malcolm | 666 Cherry St | 1997-09-09 | M | Developer |

```

```
D0002 |
| E0010 | Julia | Stiles | 777 Spruce St | 1998-10-31 | F | Developer | D0003 |
+-----+-----+-----+-----+-----+-----+-----+
+-----+
```

```
INSERT INTO Project (projNo, projName, deptNo) VALUES
('P0001', 'Web Redesign', 'D0001'),
('P0002', 'API Migration', 'D0002'),
('P0003', 'DB Optimization', 'D0003'),
('P0004', 'Automated Tests', 'D0004'),
('P0005', 'Mobile App UI', 'D0005'),
('P0006', 'AWS Server Move', 'D0006'),
('P0007', 'AI Chatbot', 'D0007'),
('P0008', 'Security Audit', 'D0008'),
('P0009', 'Helpdesk Portal', 'D0009'),
('P0010', 'VR Prototype', 'D0010');
```

```
SELECT * FROM Project;
```

```
+-----+-----+-----+
| projNo | projName | deptNo |
+-----+-----+-----+
| P0001 | Web Rede... | D0001 |
| P0002 | API Migr... | D0002 |
| P0003 | DB Optim... | D0003 |
| P0004 | Automate... | D0004 |
| P0005 | Mobile A... | D0005 |
| P0006 | AWS Serv... | D0006 |
| P0007 | AI Chatb... | D0007 |
| P0008 | Security... | D0008 |
| P0009 | Helpdesk... | D0009 |
| P0010 | VR Proto... | D0010 |
+-----+-----+-----+
```

```
INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES
('E0008', 'P0001', '2026-06-01', 30),
('E0004', 'P0001', '2026-06-01', 40),
('E0009', 'P0002', '2026-06-02', 35),
('E0005', 'P0002', '2026-06-02', 40),
('E0010', 'P0003', '2026-06-03', 20),
('E0006', 'P0003', '2026-06-03', 25),
('E0007', 'P0004', '2026-06-04', 38),
('E0001', 'P0007', '2026-06-05', 10),
```

```
('E0002', 'P0008', '2026-06-06', 15),
('E0003', 'P0006', '2026-06-07', 20);
```

```
SELECT * FROM WorksOn;
```

```
+-----+-----+-----+-----+
| empNo | projNo | dateWorked | hoursWorked |
+-----+-----+-----+-----+
| E0001 | P0007 | 2026-06-05 | 10          |
| E0002 | P0008 | 2026-06-06 | 15          |
| E0003 | P0006 | 2026-06-07 | 20          |
| E0004 | P0001 | 2026-06-01 | 40          |
| E0005 | P0002 | 2026-06-02 | 40          |
| E0006 | P0003 | 2026-06-03 | 25          |
| E0007 | P0004 | 2026-06-04 | 38          |
| E0008 | P0001 | 2026-06-01 | 30          |
| E0009 | P0002 | 2026-06-02 | 35          |
| E0010 | P0003 | 2026-06-03 | 20          |
+-----+-----+-----+-----+
```

```
SET FOREIGN_KEY_CHECKS = 1;
```

```
ALTER TABLE Employee
ADD CONSTRAINT chk_position CHECK (position IN ("Manager", "Leader",
"Developer", "Analyst"));
```

```
CREATE VIEW pManagedByF AS
SELECT p.projNo, p.projName
FROM Project p
JOIN Department d ON p.deptNo = d.deptNo
JOIN Employee e ON e.empNo = d.mgrEmpNo
WHERE e.sex = 'F' AND e.position = 'Manager'
ORDER BY p.projNo;
```

```
SELECT * FROM pManagedByF;
```

```
+-----+-----+
| projNo | projName      |
+-----+-----+
| P0001 | Web Redesign  |
| P0004 | Automated Tests |
| P0007 | AI Chatbot    |
| P0010 | VR Prototype  |
+-----+-----+
```

```
CREATE VIEW custAttributes AS
SELECT e.empNo, e.fname, e.lName, p.projName, w.hoursWorked
FROM Employee e, Project p, WorksOn w
WHERE e.empNo = w.empNo AND p.projNo = w.projNo;
```

```
SELECT * FROM custAttributes;
```

empNo	fname	lName	projName	hoursWorked
E0001	Alice	Smith	AI Chatbot	10
E0002	Bob	Jones	Security Audit	15
E0003	Charlie	Brown	AWS Server Move	20
E0004	Diana	Prince	Web Redesign	40
E0005	Evan	Wright	API Migration	40
E0006	Fiona	Gallagher	DB Optimization	25
E0007	George	Clark	Automated Tests	38
E0008	Hannah	Abbott	Web Redesign	30
E0009	Ian	Malcolm	API Migration	35
E0010	Julia	Stiles	DB Optimization	20

```
CREATE VIEW EmpProject(empNo, projNo, totalHours) AS
SELECT w.empNo, w.projNo, SUM(hoursWorked)
FROM Employee e, Project p, WorksOn w
WHERE e.empNo = w.empNo AND p.projNo = w.projNo
GROUP BY w.empNo, w.projNo;
```

```
SELECT * FROM EmpProject;
```

empNo	projNo	totalHours
E0001	P0007	10
E0004	P0001	40
E0008	P0001	30
E0002	P0008	15
E0005	P0002	40
E0009	P0002	35
E0003	P0006	20
E0006	P0003	25
E0010	P0003	20
E0007	P0004	38

```
SELECT projNo FROM EmpProject WHERE projNo = 'P0007';
```

```
+-----+  
| projNo |  
+-----+  
| P0007  |  
+-----+
```

```
SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E0001';
```

```
+-----+  
| COUNT(projNo) |  
+-----+  
| 1             |  
+-----+
```

```
SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;
```

This provides an error, since there is no aggregate function (like SUM, AVG) being used with the GROUP BY function.

7.28

In order to maintain a materialized view, each operation would need to be considered.

If a new row is being inserted into the PART table, if partCost > 1000 and the partNo is not already in ExpensiveParts, the partNo would need to be inserted into the ExpensiveParts view.

If a row is deleted, and the partCost ≤ 1000, there would be no change needed

However if it had a partCost > 1000, there would need to be a check to determine if any other row with the same partNo has a partCost > 1000. If it happens to be so, partNo stays in the view, if not it would need to be deleted from the view.